Python Loops – Practice Workbook

30 Basic-to-Medium Problem Statements for Intermediate Students

Topics: for loop, while loop, nested loops, counters, accumulators, strings, lists, and number logic.

Instructions

1. Write a separate Python program for each question.
2. Take user input wherever required.
3. Use loops to solve the problem and test with different inputs.
4. Solutions are not included so students can practise independently.

Part A – For Loop: Basic

1. Print Numbers
Print numbers from 1 to 50 using a for loop.

2. Even Numbers
Print all even numbers between 1 and 100.

3. Multiplication Table
Accept a number and print its multiplication table from 1 to 10.

4. Sum of Natural Numbers
Accept a positive integer n and calculate the sum of numbers from 1 to n.

5. Count Backwards
Print numbers from 20 down to 1 using a for loop.

6. Sum of Even Numbers
Calculate the sum of all even numbers between 1 and 100.

7. Iterate Through a String
Accept a string and print each character on a separate line.

8. Count Vowels
Accept a string and count how many vowels (a, e, i, o, u) it contains.

9. Largest Number in a List
Find the largest value in a list using a loop without using max().

10. Star Triangle
Use nested for loops to print a five-row right-angled triangle of stars.

Part B – For Loop: Medium

11. Factorial
Accept a non-negative integer and calculate its factorial using a for loop.

12. Prime Number Checker
Accept an integer and check whether it is a prime number.

13. Reverse a String
Reverse a string using a loop without using slicing such as [::-1].

14. Count Positive, Negative, and Zero
Given a list of integers, count positive numbers, negative numbers, and zeros separately.

15. Number Pattern
Use nested for loops to print these rows: 1, 12, 123, 1234, 12345.

Part C – While Loop: Basic

16. Countdown
Accept a starting number and print a countdown to 1, followed by “Time's up!”.

17. Sum Until Zero
Keep accepting numbers until the user enters 0. Display the sum of the entered numbers, excluding zero.

18. Count Digits
Accept a positive integer and count its digits using a while loop.

19. Sum of Digits
Accept an integer and calculate the sum of its digits. Example: 456 gives 15.

20. Reverse a Number
Accept a positive integer and reverse its digits. Example: 1234 becomes 4321.

Part D – While Loop: Medium

21. Palindrome Number
Check whether a number reads the same forward and backward. Example: 121 is a palindrome.

22. Guess the Number
Set a secret number and ask the user to guess repeatedly until correct. Display “Too high” or “Too low” when appropriate.

23. Password Validation
Keep asking for a password until the user enters the correct password. Then display a success message.

24. Menu-Driven Calculator
Show options for addition, subtraction, multiplication, and exit. Repeat the menu until the user chooses exit.

25. ATM Simulation
Start with ■10,000. Repeatedly offer check balance, deposit, withdraw, and exit options. Update the balance and prevent withdrawals above the available balance.

26. Multiples of a Number
Accept a number and print its first 10 multiples using a while loop.

27. Validate Input
Ask the user to enter a number from 1 to 10. Keep asking until the input is valid.

28. Fibonacci Sequence
Accept the number of terms and print that many Fibonacci sequence terms using a while loop.

29. HCF / GCD
Accept two positive integers and find their greatest common divisor using a loop.

30. Savings Goal
Repeatedly ask for a deposit amount until total savings reach or exceed ■10,000. Display the final total and number of deposits.

Student Name: _____

Date: _____ **Batch:** _____